

The 2-D Bin Packing Problem with Multiple Levels of Prioritization: A Spatial Optimization Perspective

William K. Kirschenman¹, H. Sebastian Heese², Russell E. King¹, Brandon M. McConnell¹

¹ Department of Industrial and Systems Engineering, NC State University, Raleigh, NC

² Poole College of Management, NC State University, Raleigh, NC



REFERENCES

NC STATE
UNIVERSITY

Introduction

- We introduce a **prioritization-based two-dimensional bin packing** framework.
- We **integrate ideas from facility layout planning to incorporate multiple levels of priority** – so groups of items can be placed close together, and so items/groups can be placed near bin access points for easier retrieval or off-loading.
- This approach applies in warehouse operations, military combat loading, and other contexts requiring **space efficiency and strategic item placement**.

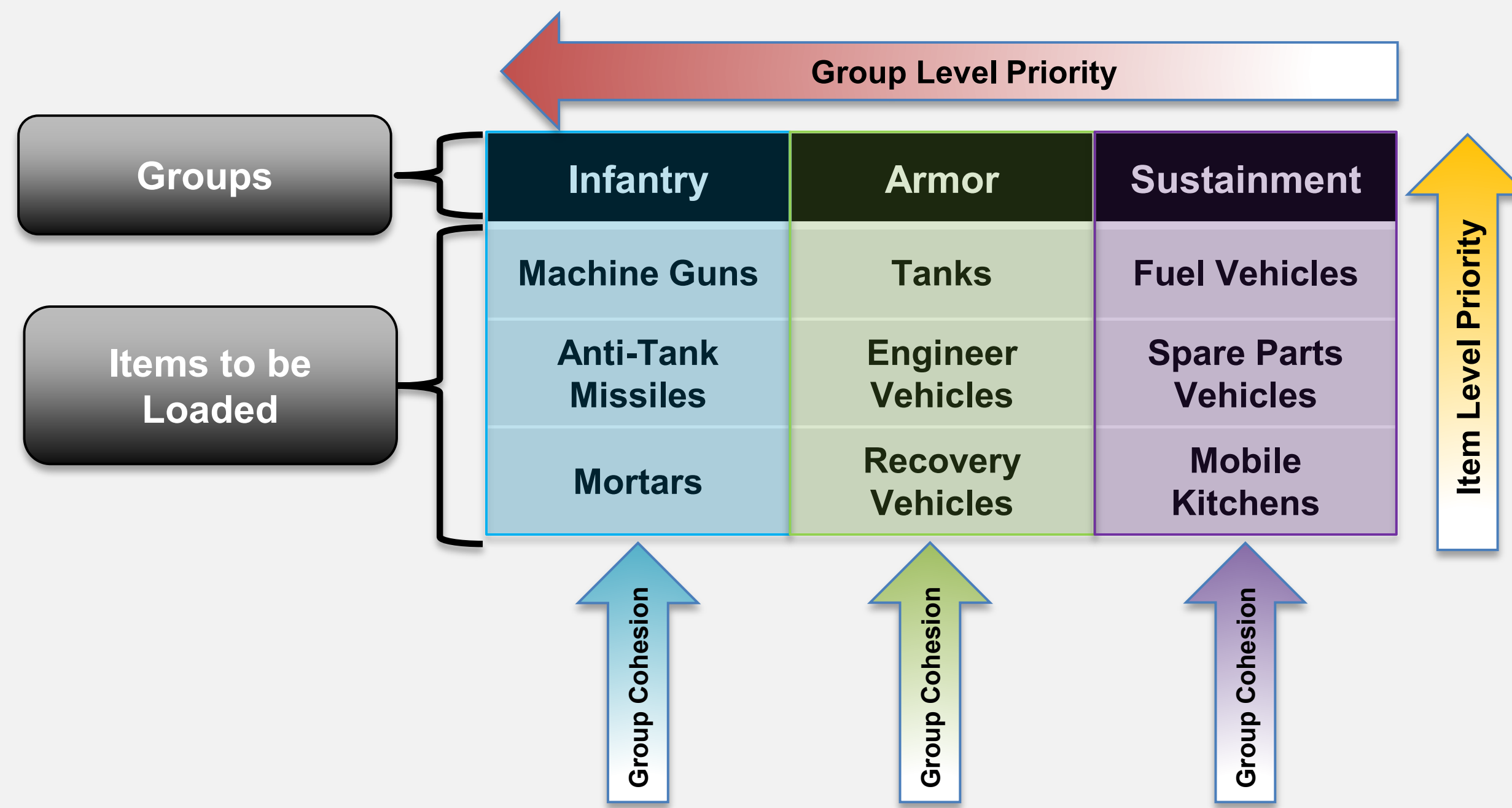


Figure 1. Example of multiple levels of prioritization and group unity in a military combat loading context.



Figure 2. Soldiers from the 1st Infantry Division off-load vehicles and equipment onto Omaha Beach at Normandy, France, at low tide during the first days of Operation Overlord in June 1944.

Literature Review

- Bin Packing:** Classical 2-D bin packing addresses geometric feasibility and minimizing unused space but often ignores nuanced distance or adjacency priorities.
- Facility Layout Planning (FLP):** FLP typically focuses on adjacency-based costs and minimizing material handling, offering frameworks for weighted distance objectives.
- Spatial Optimization:** Multi-drop container loading, load-balanced bin packing, and facility layout planning approach prioritization in different ways, but typically as constraints.

Model Formulation

Key Idea. We adapt a 2-D bin packing mixed integer linear programming (MILP) formulation to include distance-based priorities. Each item i is potentially drawn to other items ($i, k \in I$) or to a bin access point, captured by a weight matrix P .

$$\min \sum_{i \in I} \sum_{k \in I, i \leq k} P_{i,k} (d_{i,k}^x + d_{i,k}^y)$$

- $d_{i,k}^x, d_{i,k}^y$ are decision variables that measure rectilinear distances among items in the x and y dimensions (or between each item and the bin access point).
- $P_{i,k}$ is a parameter encoding group cohesion and item-to-access-point priority.
- We also **include classical packing constraints** for non-overlapping placement, orientation, and boundary feasibility.
- Our **prioritization matrix P merges item-level and group-level priorities**. **Diagonal terms** pull items toward a bin's access point. **Off-diagonal terms** encourage items in the same group to stay close.

Solution Techniques

- Single MILP Approach:** We solve the entire formulation directly with a commercial solver (Gurobi). This can prove optimal solutions for small instances but becomes intractable for larger ones.
- Groupwise Decomposition:** We fix the highest-priority group first (via a smaller MILP), then iteratively place the remaining groups in descending priority. This reduces complexity but can still be suboptimal if early placements constrain subsequent groups.
- Sliding-Window Decomposition:** We sort items by priority and solve subproblems of size w . In each subproblem, we fix the top item and introduce the next item in the list, “sliding” the window forward. This provides a good balance of runtime and solution quality.

Lower Bounds

Grid-Based Lower Bound (GB-LB)

Theorem 1. Consider a bin and items discretized into 1×1 cells, a bin access point o on the bin boundary, and n items in descending access-point priority $P_{i,o}$. For each discretized item i , let $d_{i,o}^{LB}$ be a Euclidean-based distance from o , and for each grouped pair (i, k) let $d_{i,k}^{LB}$ be a conservative pairwise distance. Then

$$LB = \sum_{i=1}^n P_{i,o} d_{i,o}^{LB} + \sum_{i < k} P_{i,k} d_{i,k}^{LB} \leq OPT$$

where OPT is the minimal rectilinear cost in any feasible packing.

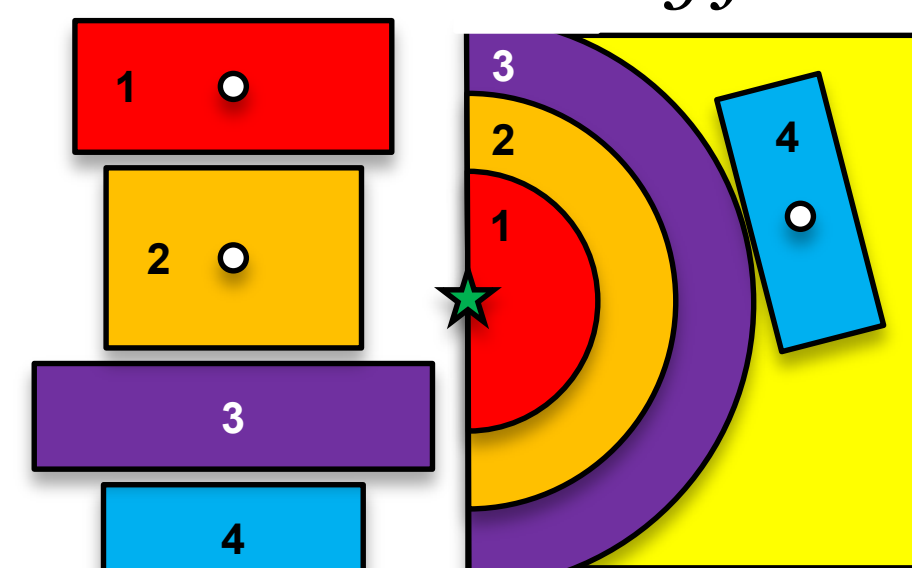


Figure 3. Conceptual example of the grid-based lower bound construction for a 4-item problem instance with a center-left bin access point.

Statistical Lower Bound

We use extreme value theory (Wilson et al., 2004) to generate multiple variant solutions, fitting a Weibull distribution to the feasible observed minima. The Golden-Alt confidence interval method produces a statistical LB on the global optimum.

Results

Groups vs. No Groups

Including group cohesion dramatically reduces pairwise-item distance costs without sacrificing costs associated with proximity to the bin access point. We compare solutions that exclude group weights (higher objective) vs. those that include group weights (tighter groups, lower total cost).



Figure 4. (Left) Visual comparison of by-group sequencing and cohesion. (Right) Objective value contributions by solving method.

Assessing Solution Quality with Lower Bounds

Table 1. Comparison of Gurobi and GB-LB for the 3 largest problem instances.

Items/Groups	Group Size	Het	Access	Obj	LB	GB-LB
52/5	m111	svhwh	CL	1.13e6	2.16e5	9.37e5
			BL	1.41e6	3.58e5	1.13e6
45/5	m111	wvsss	CL	8.58e5	1.55e5	7.30e5
			BL	1.04e6	2.75e5	8.38e5
44/5	m111	vwshh	CL	7.97e5	1.53e5	6.48e5
			BL	9.66e5	2.65e5	7.70e5

Table 2. Wilson's statistical lower bound results for the 3 largest problem instances using a center-left bin access point.

Instance	Min/Max	KS (Stat, p-val)	AD (Stat, p-val)	Golden-Alt CI	Opt. Gap (%)
42 items	6.31e5/6.74e5	(0.056/0.432)	(0.401/0.847)	(6.15e5, 6.31e5)	2.61
45 items	8.61e5/9.31e5	(0.034/0.189)	(0.117/0.999)	(8.24e5, 8.61e5)	4.58
52 items	1.13e6/1.20e6	(0.076/0.644)	(0.534/0.712)	(1.05e5, 1.13e6)	6.85

Comparative Objective Performance

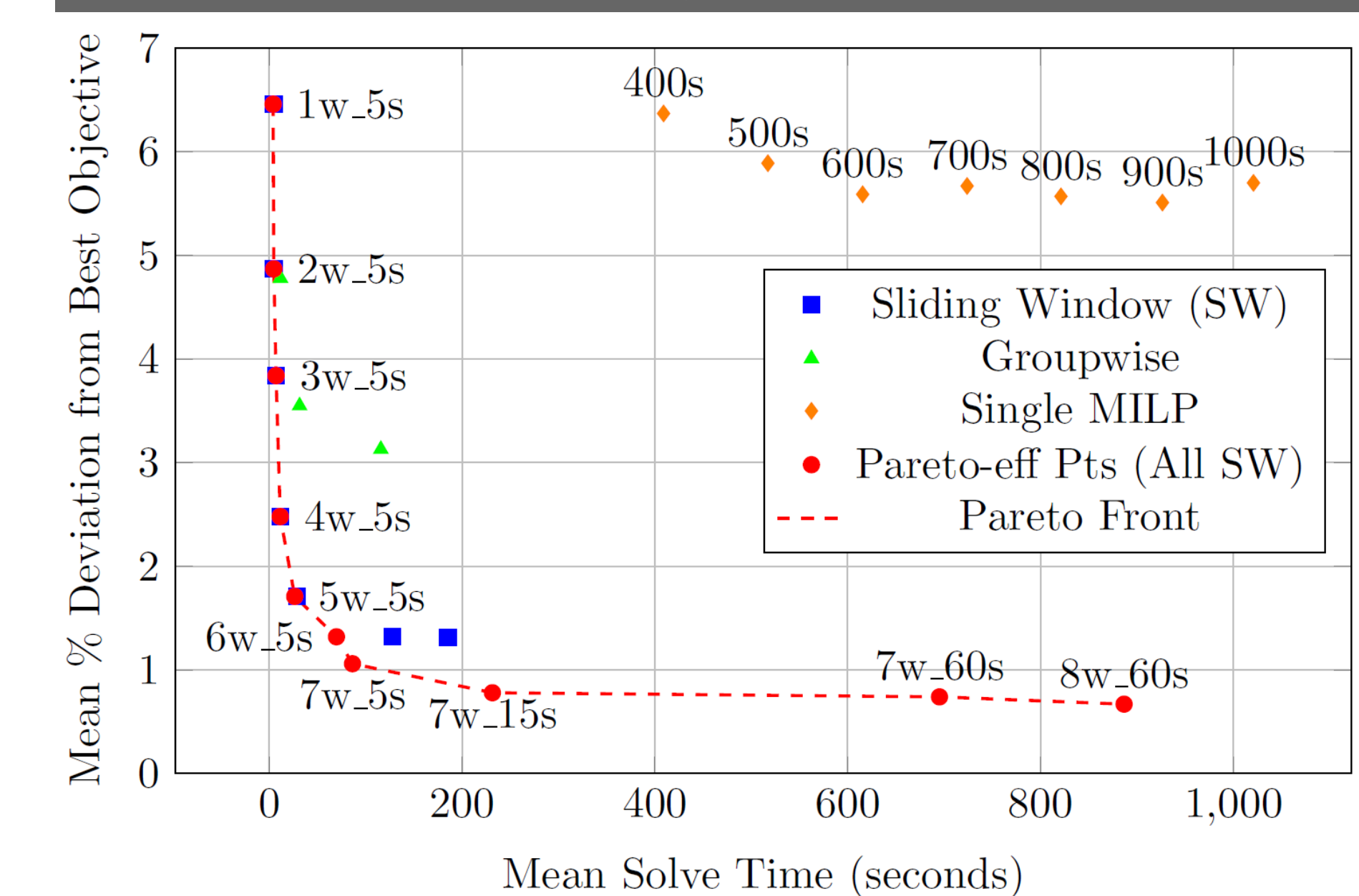


Figure 5. Pareto curve showing that sliding-window decomposition methods solely occupy the efficient frontier, providing an excellent balance between total runtime and solution quality.

Conclusion

- By **embedding a prioritization matrix in 2-D bin packing**, we handle both group cohesion and item-to-access-point proximity in one unified framework.
- Decomposition-based approaches** (groupwise or sliding window) are essential for larger instances, reducing runtime without sacrificing solution quality.
- This **methodology is broadly applicable**, from warehouse operations to combat loading scenarios where **spatial optimization** matters.